Penalizing Failures at University

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Abstract

How does increasing the cost of failing university modules affect student persistence and academic performance? We exploit a policy shock in Spain involving a two-wave increase in tuition fees—first targeting second and subsequent retake attempts, followed by a general fee increase that affected all attempts, including the first. Using a rich dataset from a very large public university, the University of Barcelona, we employ a difference-in-differences design to compare students fully exposed to the fee increases with those who received fee waivers before and after the fee changes. First, our findings show that both the penalties for module failures and the general fee increase negatively impact student persistence, by about 7 and 9 percentage points, respectively, particularly among part-time and low-performing students, raising concerns about equity. Second, from an efficiency perspective, we find evidence that financial pressure leads students to pass modules in fewer attempts, reducing the average number of attempts needed to pass by 0.03 to 0.045, although we do not observe significant improvements in grades beyond passing.

Keywords: Tuition fees, Retake costs, Higher education, Academic performance
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1 Introduction

Institutional factors within educational systems, such as incentives, play a key role in shaping students' effort and learning. Over the past decade, extensive research has examined the impact of economic incentives, particularly rewards, on students' behavior and academic performance (Angrist & Lavy, 2009; Burgess et al., 2021; Oswald & Backes-Gellner, 2014). However, insights from behavioral economics suggest that the same monetary amount can have different effects depending on whether it is framed as a reward or a penalty (see, for instance, Levitt et al. (2016)).[1] Despite these findings, relatively little is known about how negative economic incentives, such as common penalties for underperformance, affect student outcomes.

In higher education, many systems extensively use positive incentives in the form of merit-based rewards, particularly in high-tuition countries like the United States. However, the effects of negative incentives in low-tuition, publicly funded systems, such as those found in many European and Asian countries, remain under-explored. This paper seeks to address this gap by examining the impact of penalties for failures in higher education. Leveraging an unexpected increase in tuition fees for retaking failed modules[2] as a natural experiment, we empirically investigate how this negative economic incentive influences students' academic decisions and performance, shedding light on the broader implications for efficiency and equity in higher education funding policies.

More specifically, this paper examines how financial penalties for failing modules at university affect students' persistence and academic performance. To answer this question, we examine the institutional context of the Spanish public university system by exploiting a policy shock in Catalonia involving a two-wave increase in tuition fees. First, we exploit a fee increase in the academic year 2011 that targeted second and subsequent retake attempts. Second, a general increase for both first and subsequent retake attempts was introduced in academic year 2012. This context provides a unique opportunity to explore the consequences of negative incentives in a publicly funded education system with relatively low baseline tuition fees. Such systems are common in many European countries such as Germany, France, Spain, and the Nordic countries with free higher education (OECD, 2024), and in Asian contexts like Mainland China, India, and Indonesia.

With a rich dataset from a large Catalan public university, which includes detailed records on stu-

[1] The idea dates back to the prospect theory of Kahneman and Tversky (1979).

[2] In this paper, we use the term "module" to refer to complete academic units that students take as part of their university studies, such as "Microeconomics I" or "Geography of Europe", rather than the term "course", since "course" also refers to an academic year in Spanish and Catalan, which might cause confusion to some readers. Our original data use the term *assignatura* in Catalan, often translated as "subject", which is less common in the higher education context.

dent background, academic performance, and module enrollments, we analyze how students respond to these tuition fee changes. The dataset follows the cohort that entered university in 2009 through its first four academic years (2009/10 to 2012/13), covering all undergraduate programs that had adopted the Bologna curriculum by 2009, and allows us to construct a comprehensive view of student decisions every year and in every module they take, detailing persistence, course load as measured by the number of credits enrolled, and academic performance, such as grades and number of attempts until passing the module.

We use a dynamic difference-in-differences framework to compare students who were entirely exposed to the fee increases with those who were not because of fee waivers, before and after the fee increase. To account for differences between the groups that might vary the underlying trends of behavior, we further modify the strategy to allow trends to differ across background and prior performance. The analysis is carried out at both the student-year level and the student-year-module level, which allows us to estimate how financial pressure impacts not just overall persistence and course load but also student performance.

Several key findings emerge from our analysis. First, we find that penalizing module failures negatively impacts student persistence, as students who face increased tuition fees are more likely to drop out compared to those with fee waivers. This effect is more pronounced with the general fee increase, suggesting that financial pressures significantly influence students' decisions to continue their studies, decreasing persistence by about 7 percentage points when facing the retake penalty and about 9 percentage points with the general increase. Second, heterogeneity analysis reveals that the effect on persistence is driven by part-time and low-performing students, and that, under financial pressure, male students react more by dropping out compared to females.

Third, the number of credits taken by students decreases by about 5 in response to the retake penalty and by about 6 in response to the general increase. Since a typical one-semester module carries 6 ECTS credits and a full-time annual load is 60 credits, this amounts to roughly one module fewer per year. This could imply that students pass more modules and reduce their retakes, but it could also mean that students are strategically reducing their course loads to avoid failing and facing further penalties. Finally, although students exert more effort to pass modules after the fee increases, as evidenced by a reduction in the number of attempts needed to pass, we do not find significant improvements in grades beyond passing. The efficiency implication of our findings is relatively clear. Fewer resources are wasted on repeated attempts at passing modules when introducing higher retake costs. However, equality concerns exist as such fee increases push away students in disadvantaged

groups.

We further perform robustness checks by combining matching algorithms with the event studies to further ensure comparability at the student level. Overall, results show that our findings on the number of credits and performance are robust. Moreover, effects on persistence remain robust after applying re-weighting schemes and when treatment is defined on pre-reform scholarship status only, reinforcing both the efficiency and equality implications.

This study contributes to the literature in several ways. First, it contributes to the strand of literature that examines impacts of tuition fee policies. Over the past few decades, such policies have generated considerable debate. For example, England introduced tuition fees in 1998 (Murphy et al., 2019; Sá, 2019), while several German states adopted tuition fees in the early 21st century only to abolish them roughly a decade later (Bietenbeck et al., 2023; Bruckmeier and Wigger, 2014). Beneito et al. (2018) study a general tuition fee increase in Spain and find that higher fees encourage greater student effort. However, they focus on the general fee increase without separately examining the effects of attempt-specific fees. Most existing research considers tuition fees as a whole, yielding mixed results on their impact. The study by Garibaldi et al. (2012) is one of the few exceptions that focus on the structure of tuition fees, leveraging a scheme in an Italian university where fees were a step function of family income. Compared to this literature, this paper is the first to disentangle the implications for student persistence and effort of a general fee increase versus an attempt-specific fee.

Second, by emphasizing the retake costs among other components of tuition fees, we also contribute to the literature on retake examinations. Existing research mainly focuses on retaking (or resitting) exams (Bietenbeck et al., 2023), particularly high-stakes exams (Krishna et al., 2018; Nijenkamp et al., 2016), rather than module retakes. Recently, Bertola (2024) examines a setting in which students are allowed multiple exam resits and concludes that such opportunities not only decrease the reliability of results as indicators of ability but also unnecessarily increase related costs. Retaking entire modules after exhausting all resit opportunities is also a common but under-explored aspect of higher education. Our paper contributes by examining module retakes and linking them to costs and financial incentives.

Third, our paper also relates to the studies of student incentives. This literature mainly focuses on school education and has shown that financial incentives can positively impact student outcomes, including both education completion and academic performance (Angrist and Lavy, 2009; Bettinger, 2012; Fryer, 2011). Recently, there has been some research examining financial incentives in the context of higher education, mainly using randomized controlled trials. For example, Leuven et al. (2010)

run an experiment that stimulates freshmen to exert more effort and find diverging effects for high- and low-performing students. Angrist et al. (2014) provide performance-based financial rewards to students at a Canadian community college, but only finds at most modest effects on improving achievement. While these studies focus mainly on positive incentives, we contribute to the literature by examining students' reactions to negative incentives using observational evidence at a larger scale, as compared to experiments that often involve only hundreds of students.

The rest of the article is organized as follows. Section 2 presents the fee structure for public universities in Spain and zooms in on the policy shock that introduces the penalization on module failures. Section 3 describes the data we use and summarizes the sample for the analysis. Section 4 introduces the empirical strategy for identifying the effects of increasing retake costs. In Section 5 we present the results of the main analysis, dividing these into behavioral responses at the student-year level, and the effects on academic performance at the student-year-module level. Section 6 further examines heterogeneous effects on persistence across groups to examine the equity concerns of introducing a financial penalization. Finally, Section 7 concludes.

2 Institutional Background

2.1 Tuition Fees and Fee Waivers

University fees that students pay in Spanish public universities can be decomposed into two parts, tuition fees and administrative service costs. Tuition fees are usually the dominant part and are paid according to the number of credits a student enrolls, and per-credit rates vary across study fields and number of attempts. Administrative service costs include basic services such as registration fees and compulsory student insurance, and are charged at a uniform rate, as well as additional services including optional insurance, photocopying, and sport facilities which students can choose whether to use or not.

If a student secures a scholarship, they will receive a fee waiver that helps cover their tuition fees. According to the Spanish Law that regulates scholarships and student aid (*Real Decreto 1721/2007*), the student has to meet both economic and academic requirements to be eligible for the scholarship. The scholarship is a package that includes various components, including cash allowance and compensations for living, traveling, study materials, matriculation, etc. For our study, the most relevant part is the "compensation for matriculation", or fee waiver, which means that students are exempt from paying tuition fees.

Economic requirements. As stated in Article 13 of *Real Decreto 1721/2007*, scholarships cannot be awarded to applicants whose family income and assets exceed a certain threshold, which is calculated according to the number of family members and family income. Eligible students are divided into several categories. All students who are awarded the scholarship receive the fee waiver, the most basic component. However, as family income worsens, students might be eligible to receive an additional income allowance. The criteria that determine the student's financial situation include both the family's assets and income.[3]

Academic requirements. The academic threshold includes conditions on both prior performance and matriculation status. If a student does not pass at least 80% of the credits enrolled in the previous year (60% in engineering and architecture), they will not be awarded the scholarship (Article 22).[4] It is worth noting that the law also requires students to enroll in at least 60 European Credit Transfer and Accumulation System (ECTS) credits per academic year to be able to apply for the scholarship (or 30 credits in the first semester, when applying). However, from academic year 2009, students are also eligible for certain components of the scholarship, including the fee waiver, if they choose to enroll for 30 to 59 credits, as referred to as "partial matriculation" in the regulation.[5]

2.2 The Retake Fee Shock

Under Spain's decentralized education system, the central government devolves the right to set fee levels to regional governments. The policy shock we investigate in this paper happened in Catalonia.[6] In 2011 and 2012, tuition fees for public Catalan universities increased. In the first wave of fee rise in 2011, only fees for retaking the same modules that students had failed twice in previous years changed. In 2012, while retake costs continued to rise, a general increase for all attempts, including the first, was further introduced. However, before and after this period, university tuition fees in the region remained relatively stable, providing us with an ideal policy experiment to understand the impacts of raising module-specific attempt fees.

[3]For example, according to *Real Decreto 557/2010*, regardless of family income, if the value of urban residential real estate assets held by the family exceeds 42,900 euros, the student will not be eligible to receive the scholarship. Different family income categories are also specified in the law. For instance, to be eligible for the fee waiver component of the scholarship, the family shall not earn more than 34,352 euros per year if the student comes from a three-people household, and not more than 40,796 euros if the family has four members.

[4]These thresholds applied to the scholarships for the academic years 2009/10 to 2011/12. For 2012/13, Real Decreto 1000/2012, published in July 2012, raised them to 90% in arts and humanities and in social and legal sciences and to 65% in engineering and architecture, and kept them at 80% in sciences and health sciences.

[5]See also *Real Decreto 922/2009* and *Real Decreto 557/2010*, which add this to the original regulation, *Real Decreto 1721/2007*.

[6]Catalonia is the region with the second-highest GDP, after Madrid, accounting for 19% of Spain's total GDP. With a population of 8.2 million inhabitants, representing 16.5% of the Spanish population, it is also the second-largest region in demographic terms, after Andalusia.

The key consequence of these two waves of tuition fee changes is that they altered the structure of the fee scheme, greatly widening the differences in per-credit prices for different numbers of attempts made to the same module. Before 2010, in most regions of Spain, and especially in Catalonia, the levels of tuition fees for enrolling in a module for the first time and for subsequent retakes of the same module were different, but the gaps were relatively small, as shown in Figure 1. With the new policy, in 2011, the per-credit price of the second and subsequent retakes rose sharply relative to the previous year, while the prices of the first two attempts did not change discontinuously. Figure 1 presents the magnitude of changes for Humanities and some majors in Social Sciences.[7] In 2011, the cost per credit for the second retake nearly doubled, jumping from 25.36 euros in the previous year to 45.48 euros. In 2012, the per-credit price for the second retake further increased by 20.25 euros, and a new divergence was created between the second retake and third and subsequent attempts, with the per-credit prices for the first attempt and first retake also raised by about 10 euros.

Importantly for our identification strategy, in Catalonia the law that regulates tuition fees per credit and degree program is typically published in July for the coming academic year. This information is meant to be public and available to students from its publication date. Therefore, by the time the law is announced, students have already finished their final exams (these take place in June). Besides, students pay for the modules enrolled for the following academic year when it starts in September. Therefore, from the perspective of students, the information comes after the realization of the decision of how much effort to invest and how many credits to take in the previous academic year, but before that of the next academic year. This likely rules out strategic reactions in the final exams of the previous academic year due to anticipation of the fee changes in the next.

3 Data

We use administrative records from the University of Barcelona (UB), the largest university in Catalonia.[8] Our data contains module-level and student-level information. At the individual level, we observe academic records on study program and graduation, as well as fee status every year. Importantly, we also have detailed module-level exam records for all the modules a student takes during their undergraduate studies. These records are linked to student background characteristics registered upon enrollment at the university, including college entrance exam (PAU) results, high school attended,

[7] The evolution patterns of the fees for natural sciences, health sciences, and engineering are identical to the one presented in this graph, with their fee levels higher. Therefore, for brevity, we show the graph only for Humanities and Social Sciences.

[8] According to the Statistical Institute of Catalonia (https://www.idescat.cat/indicadors/?id=aec&n=15731&tema=educa&t=200100), since 2000, 30% of all public university students have attended UB. This represents 20% of all students in higher education, making it the largest university in the region.

foreign language abilities, parental education and occupation.

Using the information on student records, we first construct a student-year balanced panel detailing enrollment status at university, number of credits enrolled, scholarship status, and fee status over the four years of study. We combine this panel of academic and fee records with the student's background characteristics. For a detailed analysis of student performance, we also use a panel at the student-year-module level. These records include the modules a student takes at a given academic year, the number of credits the module carries, whether it is compulsory, whether they attend the final exam, and the grades they obtain. Using the module identifiers, we further construct variables on the number of attempts a student makes associated with each record and how many attempts they need to pass.

Our dataset covers the universe of students enrolled in all undergraduate programs that had already implemented the Bologna reform in 2009.[9] For our main analysis, we therefore consider the cohort that started university in September 2009. For this cohort, the tuition fees increased when they were about to start their third year of study. Therefore, we are able to observe their behavior and performance for the first two years without the increased fees for retakes, and for another two years after the introduction of the tuition penalization on failures. To analyze the academic records of the modules, we only consider performance in compulsory subjects, since compulsory modules cannot be strategically avoided or replaced once students fail to pass.

An important feature of our setting is that scholarship status can change from year to year. Table 1 presents year-to-year transition matrices for fee-waiver status in our sample. The fee-waiver status is highly persistent: among self-funded students, 92–94% remain self-funded in the following year, and among fee-waiver recipients, 71–80% retain their waiver. Overall, 74.1% of students maintain the same status across all four years of study (60.6% are always self-funded, and 13.4% always hold a fee waiver), while 25.9% experience at least one status change. Importantly, the transition rates are broadly similar across the three year-to-year transitions, suggesting that the policy change did not itself trigger systematic changes in scholarship status. Operationally, we classify a student as a fee-waiver student if she obtained a fee waiver throughout the first three years of study. We define treatment based on the first three years rather than all four to avoid potential endogeneity from scholarship status changes that could be induced by the fee shock itself in the fourth year.

[9] In academic years 2009 and 2010, as part of the Bologna process, there were universal curriculum reforms happening in Catalonia (see Appendix B for detailed information). By focusing on the undergraduate programs that had already implemented the reform in academic year 2009, we avoid any confounding effects from its implementation, since all students are observed under the same post-Bologna scenario.

Table 2 reports the summary statistics for the analytical sample by whether the fees were waived, and Appendix Table A12 reports normalized differences for all pre-determined characteristics. The analytical sample consists of 3,814 students, 603 with a fee waiver and 3,211 without. One self-funding student is enrolled in two degree programs and enters the panel once for each, so the student-year panel contains 3,815 student-program units and 15,260 observations. One further self-funding student lacks information on parental background and is therefore not included in the specifications with background-specific trends (15,256 observations). As expected, students with a fee waiver (Column 1) come from more disadvantaged families compared to students without a fee waiver (Column 2). Their parents are less likely to have higher education, more likely to be unskilled workers, and more likely to be immigrants. The average fees paid per year for students with the fee waiver are about 125 euros, compared to above 1,000 for those without a fee waiver. We also see that students with a fee waiver tend to stay on in university with a very high persistence, close to 100%, compared to 95% for students without the fee waiver. Students with a fee waiver enroll in 60 credits per year on average, which corresponds to a full-time load (a four-year degree requires 240 ECTS credits).

4 Identification Strategy

Our goal is to use the tuition fee shocks to identify the effects on students' persistence and effort. As shown in Section 2, in 2011, the rise in tuition fees only happened to the second and higher retakes, which offers us the opportunity to separate the effects of failure penalization without being concerned with an increase in the baseline tuition fees. We further use the effects of the second wave of increase in 2012 as additional evidence on the impact of tuition fee increases.

In order to identify these effects, we use students who have to pay tuition fees as the treated group since they are sensitive to fee changes, and compare them to a group of students who receive scholarships during their studies in university and thus are exempt from paying tuition fees. Operationally, we classify a student as control ($T_i = 0$) if they held a fee waiver in each of the first three years of study, and as treated ($T_i = 1$) otherwise, that is, if they paid tuition fees in at least one of these years. Recall that we define treatment based on the first three years rather than all four to avoid potential endogeneity from scholarship status changes that could be induced by the fee shock itself in the fourth year. Because holding a fee waiver in the third year requires being enrolled in that year, control students under this definition persist through the third year by construction. We therefore also report estimates based on a definition that uses only pre-reform information, in which the control group consists of students who held a fee waiver in both of the first two years (Appendix Tables A6

and A7). We use the data for the 2009 cohort, who were in the middle of their studies when the penalizations were introduced. Specifically, the higher retake cost for the third and subsequent attempts was introduced in their third year of study, whereas in their fourth year of study both the costs for retakes and the first attempt experienced an increase. Appendix Table A8 reports the corresponding changes in total fees paid by self-funding students relative to fee-waiver students: they rise substantially in the fourth year in all specifications, whereas in the third year, when only repeated enrollments became more expensive, they change little in most specifications, in part because treated students also enrolled in fewer credits.

We begin with the analysis at the student-year level, examining the effects on student behavior: persistence and credits enrolled. Persistence is defined as a binary variable taking the value of one if the student stays enrolled at UB and zero if a student drops out. The other outcome of interest at this level is the number of credits a student enrolls in, which reflects the strategic reaction of a student when facing the penalization, since they might choose to enroll in fewer credits to ensure a higher probability of passing. We set up the following specification to estimate the effects at this level:

$$y_{it} = \alpha_i + \mu_t + \sum_{\substack{s=1 \\ s\neq 2}}^{4} \beta_s (T_i \cdot YS_s) + \sum_{k=1}^{K} \sum_{\substack{s=1 \\ s\neq 2}}^{4} \gamma_{ks} (X_{ik} \cdot YS_s) + \varepsilon_{it}, \tag{1}$$

where y_{it} is the outcome of interest of student i at year of study t (from 1st to 4th), α_i and μ_t are student and year of study fixed effects, and ε_{it} is the error term. T_i indicates whether the student is in the treated group, and YS_s is an indicator equal to one if the observation corresponds to year of study s. In our setting, the policy change happened in the summer after the second year of study for the 2009 cohort. Therefore, we use the second year of study as the reference point of the event study and omit it in the estimation.

We have shown in Table 2 that treated and control individuals differ in terms of observable time-invariant characteristics. This does not pose a problem for our identification strategy since all potential (observed and unobserved) time-invariant characteristics are absorbed by the inclusion of α_i. However, individuals with different characteristics could evolve differently over time, which could confound the fee policy shock. We include characteristic-specific time trends ($X_{ik} \cdot YS_s$) in order to address this concern. X_{ik} includes parental education and occupation, the student's admission grades to the program of study, her English and French language abilities, major, gender, and whether the student is from an immigrant family. Major enters as a set of indicators, so that each degree program has its

own year-of-study effects. In some specifications, we also include the students' performance before the increase, measured as the GPA the year before the shock, in order to allow for more flexible trends. Thus, the identification relies on the following modified parallel trends assumption: In the absence of the shock, outcomes for treated and control students who share the same characteristics would have followed parallel trends over time. We further test whether outcomes differed significantly between treated and control groups in period 1. Unfortunately, given the short duration of our study period (four years), we cannot precisely test for parallel trends over a longer horizon.

To further investigate the impact of penalizing module failures on students' academic outcomes, we use detailed records of student achievement for every module taken. We adjust Equation 1 in the following way:

$$y_{imt} = \alpha_i + \mu_t + \delta_m + \sum_{\substack{s=1 \\ s\neq 2}}^{4} \beta_s (T_i \cdot YS_s) + \sum_{k=1}^{K} \sum_{\substack{s=1 \\ s\neq 2}}^{4} \gamma_{ks} (X_{ik} \cdot YS_s) + \varepsilon_{imt}, \tag{2}$$

where m indexes modules and δ_m denotes module fixed effects.

5 Results: Student Responses to the Fee Increase

5.1 Effects on Student Behavior

We first examine the effects on student persistence and the number of credits taken. The analysis at this level is restricted to students who were enrolled in the first two years, so that both groups consist of students who were actively enrolled when the fee changes took effect.

In terms of persistence, it is reasonable to expect that some students might choose to drop out of university if the financial costs of studying are high, as is shown in previous research (e.g., Bound and Turner, 2011; Castleman and Long, 2016). The effects on persistence are presented in Table 3. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student fixed effects. Column (3) incorporates group-specific time trends based on background characteristics, and Column (4) further includes group-specific trends on students' performance before the increase. As is noted above, since we only include students that do not drop out in the first two years, by construction, there is no variation in the first year of study for persistence, and thus we cannot report the estimate of this period for this outcome.

The results in Table 3 show that students facing the increase in attempt-specific fees in their third

year of study are between 6.6 and 8.4 percentage points more likely to drop out, compared to those with a fee waiver. In the fourth year of study, with an additional general increase in tuition fees for all attempts, the effects are larger in magnitude, showing a drop in persistence of between 8.6 and 10.8 percentage points. Because the sample includes only students enrolled in the first two years, persistence equals one for all students in the base period, and the estimates capture the difference between treated and control students in continuation from the third year onwards. These effects are economically significant, as compared to an unconditional mean for the probability of staying on in university in the analytical sample of 95.8%. We plot the results of our preferred specification (Column 4) in Panel A of Figure 2.

We now turn to the number of credits taken by students. Facing increasing tuition fees, students are predicted to take fewer credits for at least two reasons. First, they might take fewer credits at a given academic semester, so that they can focus on less material to ensure that they pass and are not charged higher fees on the following attempt. Second, especially in the fourth year, after experiencing the extra penalty for failing a module in the previous year, students might pass more modules and, as a result, take fewer credits due to fewer retakes. Table 4 shows that, prior to the policy introduction, there were no significant differences between treated and control students in the number of credits taken during the first year of study. However, by their third year of study, when the penalization was first introduced, students without a fee waiver significantly reduce the number of credits taken, by between 4.5 and 4.8 credits, depending on the specification. This reduction in the number of credits taken upon the policy introduction would support the first channel described above. In the fourth year, when facing a general increase in fees for both first and subsequent retake attempts, students further significantly reduce the amount of credits taken, by between 6.2 and 7.3 ECTS depending on the specification used. Again, results are very similar across specifications, and we plot the results of our preferred specification in Column 4 in Panel B of Figure 2. These results are economically significant as, on average, a one-semester module carries 6 credits. This implies that students took about one module fewer in their fourth year of study. As we show in Section 5.3, this reduction combines students who leave the university, who no longer enroll in any credits, with a smaller reduction in the course load of students who remain enrolled.

To further check the robustness of the analysis, we conduct propensity score matching (PSM) based on student characteristics, before regressing the same event study specification. For persistence, we first report the robustness checks in Table A1. We first repeat the specifications for the main analysis in Table 3 using the sample obtained from PSM with the Epanechnikov kernel, as reported in the first four columns, and then Column 5 shows the results after applying the matching weights in the

regression with individual fixed effects. The common support excludes 259 students, including all students in six degree programs that have no fee-waiver recipients, and the estimates on this sample in Columns (1)–(4) are very close to those in Table 3. Overall, we observe similar estimates for all the columns, with the effects remaining highly significant across all specifications including after re-weighting. Further, in Panel A of Table A2 we report the results applying various PSM algorithms[10]. We confirm that these results remain robust across different matching algorithms. Table A3 shows the estimates using the sample on the common support obtained from the Epanechnikov kernel matching, and results are similar to those reported in Table 4. Further analyses with various PSM algorithms are reported in Panel B of Table A2, yielding robust results. Finally, we re-estimate the student-level models defining treatment on pre-reform scholarship status only, so that control students are not required to be enrolled in the third year (Appendix Tables A6 and A7). The estimated effects are somewhat smaller but remain statistically significant: persistence falls by 4.5 and 6.8 percentage points in the third and fourth years, and the number of credits enrolled by 3.0 and 4.3, respectively.

5.2 Effects on Academic Performance

The analysis at the student-year level reveals that higher tuition fees negatively impact persistence and reduce the number of credits students take. With higher retake costs and a fixed amount of credits to pass, a rational choice would be to exert more effort to pass and avoid being penalized. This holds true in general even in their fourth year of study when the fee increase is general, because the fees for subsequent attempts after failure all rise. We first examine the effects on the average number of attempts a student needs to pass a module in Panel A of Table 5.[11] In our preferred specification (Column 4), there are no significant differences before the fee change. After the introduction of the penalty, the average number of attempts a student needs to pass a module significantly decreases by about 0.03 and reaches 0.045 in the fourth year, or about 2.5 and 3.8 percent of the pre-treatment mean of 1.2 attempts. Without student and module fixed effects (Column 1), the estimates also reflect differences in the modules taken by the two groups, as the large first-year coefficient indicates, and the third-year coefficient is positive; once the comparison is made within students and modules (Columns 2–4), the number of attempts falls in both post-reform years.

Further, we zoom in on the probability of passing a module within two attempts, an outcome

[10] We show results with kernel matching, radius matching and k-nearest neighbors (kNN) matching algorithms, each with several variants. For kernel matching, we report results with the Epanechnikov kernel, the Gaussian kernel, and the biweight kernel. For radius matching, the radii we use include 0.05, 0.01, and 0.1. For kNN, we use both 3 and 5 nearest neighbors. In general, all results are similar.

[11] This is possible because we are able to use unique module identifiers to trace students' modules taken until successfully passed.

targeting the effects of retake penalization. Panel B of Table 5 shows that students are about 0.7 percentage points more likely to pass the module within two attempts in both the third and fourth years. In terms of magnitude, the pre-treatment mean of 96.5% implies that 3.5% of module enrollments require more than two attempts, or about 0.35 modules per student and year for a typical load of 10 modules. An increase of 0.7 percentage points in the probability of passing within two attempts therefore corresponds to a reduction of about one-fifth in the share of modules that require three or more attempts. Table 5 presents the results for compulsory modules.[12]

Upon passing, we are also interested in examining whether higher costs, and especially retake penalization, can induce higher levels of student effort in terms of higher grades. Table 6 presents the results using the 10-scale numeric grades as the outcome variable. Results show that the two groups do not behave significantly differently before the fee change. Overall, with the increased fees, we also do not find any effects of students improving their grades conditional on passing.[13] The confidence intervals of the Lee bounds in Appendix Table A11 rule out improvements larger than about 0.2 grade points on the 10-point scale. Figure 3 summarises the event-study estimates for both outcomes at once, confirming that the significant reduction in attempts emerges in Year 3 and deepens in Year 4, while the grade effect remains close to zero throughout.

5.3 Graduation and Attrition

A natural follow-up question is whether the higher dropout rates among self-funding students documented in Section 5.1 translate into worse graduation outcomes. To examine this, we complement the panel event-study analysis with cross-sectional regressions of on-time graduation (graduating within four years) on treatment status, progressively controlling for background characteristics, major fixed effects, and pre-treatment academic performance (Tables A9 and A10 in the Appendix). The unconditional gap in on-time graduation between treated and control groups is 30.4 percentage points. However, this gap shrinks to 5 percentage points once pre-treatment academic performance is controlled for, although it remains statistically significant, suggesting that most of the raw graduation gap reflects pre-existing differences in academic preparation.

When restricting the sample to students who remained enrolled for all four years in Table A10,

[12] Including the small number of module records that correspond to cancelled enrollments or to credits obtained through recognition or validation (less than 1% of records) leaves the results unchanged; see Appendix Tables A4 and A5.

[13] Given that this null effect could indicate either that students do not exert extra effort once they pass the module, or that more students pass and pile up in the lower bands, which masks the effects of an overall increase in performance, we further test whether introducing higher costs targeting both second retake and all attempts changes the distribution of grades conditional on passing. However, we do not find any credible evidence of such an effect. Detailed results are available upon request.

the treated coefficient becomes statistically insignificant in the fully controlled specification, indicating that, conditional on prior performance, stayers in the two groups graduate on time at similar rates. A back-of-envelope decomposition suggests that excess dropout accounts for about 24% of the total graduation gap, with the remaining 76% reflecting differences among stayers that are almost entirely attributable to selection on pre-treatment characteristics.[14] These results suggest that the fee policy's impact on graduation operates primarily through the persistence channel documented above, rather than by additionally impairing the academic progression of students who remain enrolled.

Furthermore, since the treated group exhibits substantially higher dropout rates than the control group, we further decompose the effects along the extensive and intensive margins and examine whether the outcomes that are only observed for students who remain enrolled are affected by differential selection (Appendix Table A11). For credits enrolled, the reduction documented in Section 5.1 operates along both margins: students who leave the university no longer enroll in any credits, and students who remain enrolled reduce their course load by 0.8 credits in the third year and 1.5 credits in the fourth, which accounts for roughly 15 and 20 percent of the total effect, respectively. To assess differential selection, we compute Lee (2009) bounds: we trim the control group by the excess share of treated students whose outcome is not observed (between 8.4% and 12.5%, depending on the outcome and year), dropping either the control students with the smallest or those with the largest changes in the outcome, and re-estimate our preferred specification on each trimmed sample. The reduction in the number of attempts needed to pass is robust to this procedure: the bounds are $[-0.033, -0.010]$ in Year 3 and $[-0.053, -0.017]$ in Year 4, and the conservative 95% confidence interval excludes zero in Year 4. The bounds for the probability of passing within two attempts remain non-negative in both years, although their confidence intervals include zero, and grades conditional on passing remain indistinguishable from zero. For credits enrolled by students who remain enrolled, the bounds are $[-1.74, 0.10]$ in Year 3 and $[-3.22, -0.48]$ in Year 4.

In summary, our analysis highlights the significant influence of financial pressures on student decision-making, revealing that higher tuition fees negatively impact persistence and reduce the number of credits students take. Facing higher costs, students also tend to reduce the number of attempts needed to pass modules, with further evidence suggesting that students focus primarily on avoiding penalties rather than improving grades after passing. The graduation analysis reveals that the fee

[14] Since dropouts mechanically count as not graduating, each group's on-time graduation rate equals its stayer share times the stayer graduation rate. The treated group's dropout rate is 11.9%, versus 0.7% for the control group. In a counterfactual where the treated group's dropout rate is set equal to the control group's, but the stayer graduation rate is held fixed at its observed value (66.4%), the treated group's overall graduation rate would rise from 58.7% to 66.0%, reducing the total gap from 30.4 pp to 23.1 pp. The 7.3pp reduction attributable to excess dropout thus accounts for $7.3/30.4 \approx 24\%$ of the total gap.

policy's effect on degree completion operates primarily through the dropout channel, and Lee bounds show that the reduction in the number of attempts needed to pass is robust to the resulting differential attrition. Therefore, the efficiency implications are relatively clear, as fewer resources are wasted on repeated attempts at passing modules. However, the equality implications are less straightforward. To further explore how financial pressures might be pushing away certain groups of students, the next section focuses on identifying those who are most at risk of being disadvantaged by these policies and how their persistence is affected.

6 Who are Being Pushed Away?

To investigate potential heterogeneous effects on persistence, we adjust Equation 1 in Section 4 to include possible sources of heterogeneity. Theoretically, the effects are likely to be driven by students who are in a more difficult financial situation and facing a higher threat of being penalized due to poor performance. Therefore, we examine the impacts of socio-economic status and study mode, as well as the effects on marginal performers. Prior studies have provided mixed evidence on whether male or female students are more sensitive to incentives (e.g., Angrist et al., 2009; Levitt et al., 2016; Rodríguez-Planas, 2012). Thus, we are interested in examining whether there exist any different effects between male and female students in our setting.

Specifically, we modify the previous specification (Equation 1) to incorporate interaction terms that allow for heterogeneous effects of the fee policy shocks according to student characteristics. The regression model then becomes the following triple difference specification:

$$y_{it} = \alpha_i + \mu_t + \sum_{\substack{s=1 \\ s\neq 2}}^{4} \beta_s (T_i \cdot YS_s) + \sum_{\substack{s=1 \\ s\neq 2}}^{4} \theta_s (D_i \cdot YS_s) + \sum_{\substack{s=1 \\ s\neq 2}}^{4} b_s (T_i \cdot YS_s \cdot D_i) + \sum_{k=1}^{K} \sum_{\substack{s=1 \\ s\neq 2}}^{4} \gamma_{ks} (X_{ik} \cdot YS_s) + \varepsilon_{it}, \quad (3)$$

where D_i is a time-invariant indicator for the subgroup of interest. The terms $D_i \cdot YS_s$ allow the subgroup to follow its own path over the years of study, so that b_s captures the additional effect of the fee increase on treated students in the subgroup rather than differences in trends between subgroups. As in Section 5.1, by construction, the first-year estimates cannot be reported here as a result of the restriction to include only students who are present in the first two years, to ensure comparability between treated and control groups.

Figure 4 presents heterogeneous effects on persistence across all four groups; full estimates are reported in Appendix Table A13. We construct a proxy for low socio-economic status (SES) based on paternal education and occupation.[15] We find that students from low-SES backgrounds experience an additional dropout effect of about 3 percentage points in both the third and fourth years, although this estimate is not statistically significant once background-specific trends are included. Since direct household income data are not available in administrative records, parental background serves as an indirect proxy for financial circumstances. To complement this, we use a second indicator that is likely to reflect binding financial and time constraints in our context: whether the student was studying part-time in the first and second years according to the number of credits enrolled. The idea is that, unless they are in a very demanding situation, students normally do not choose to enroll only partially, as in those years modules are in general compulsory and a prerequisite for later modules. We find that part-time students are the ones most strongly driving the effects. In particular, in the third year when only retake penalization applies, part-time students face an additional dropout effect of about 20 percentage points, and the difference further increases in the fourth year to about 28 percentage points.

When it comes to gender differences, we find that female students are significantly less likely to drop out when facing higher costs, with the difference ranging from about 2 to 6 percentage points; only the smallest of these estimates, for the third year in the matched sample, is not statistically significant. Part of the reason is that males in general are more likely to be low performers compared to females, which leads us to the last source of heterogeneity presented in Figure 4. The heterogeneity variable, D_i, is a dummy indicating whether the student performed marginally in the year before the shock. We define a marginal performer as one with a GPA lower than 6 in the year before the fee rise, but the results stay robust when we use a threshold of 5.5. Results show that marginal performers are the ones driving the effects on persistence in both waves of fee increase, and the difference is about 16 percentage points in the third year with only the retake penalization and about 20 percentage points in the fourth year with the general fee increase for both retake and first attempts.

Based on these four groups of heterogeneity analyses, we further examine what drives the concentration of effects among low performers. Figure 5 shows the further decomposition of this source of heterogeneity through two complementary lenses. Panel A provides a descriptive comparison of group composition between the low-performer subgroup and the full sample. The share of low-SES students is nearly identical across the two groups, 23.2% and 22.8%, respectively, suggesting that

[15] Specifically, we classify a student as low SES if the father's highest educational attainment is below tertiary level (i.e., secondary education or below) and the father's occupation falls outside managerial, professional, or skilled industrial and service worker categories. This effectively captures families where the father holds a manual, agricultural, or unskilled occupation without post-secondary education. Results are robust to a broader definition that retains skilled industrial workers in the low-SES group.

low performers are not disproportionately drawn from socio-economically disadvantaged families. By contrast, the share of male students is markedly higher among low performers (36.1%) than in the full sample (29.1%), indicating that males are over-represented in this academically marginal group. Panel B then examines whether, conditional on being a low performer, belonging to either subgroup further amplifies the dropout response. Within the low-performer subsample, male students face an additional dropout effect of approximately 5 percentage points in Year 3 and 9 percentage points in Year 4. These estimates are imprecise: with individual fixed effects they are statistically significant at the 10% level in Year 3 and at the 5% level in Year 4, but not once background trends are added. The corresponding interaction for low-SES students within low performers is smaller in magnitude and not statistically significant. Taken together, these patterns suggest that the gender composition of low performers accounts for at most a small part of their larger dropout response; academic standing itself remains the main margin along which the fee increases operate.

Therefore, our analysis suggests that the introduction of penalization for retakes and the subsequent fee increases had heterogeneous effects, disproportionately impacting students who are already vulnerable. Specifically, the effects are concentrated among students who study part-time or performed poorly in the year prior to the increase, while students from lower socio-economic backgrounds experience a smaller additional effect, which is not statistically significant once background-specific trends are included. These patterns are consistent with binding financial and time constraints, although our data do not allow us to separate the two. Female students show a significantly lower likelihood of dropping out under increased financial pressure compared to males. These findings highlight that fee increases and financial penalties can have unintended consequences, potentially exacerbating inequalities and limiting access for students who are already disadvantaged.

7 Conclusion

This paper set out to investigate the effects of financial penalties for module failures on student outcomes in higher education, using a natural experiment involving tuition fee increases in Catalonia. Our analysis leveraged two waves of tuition fee increases, one specifically targeting retake attempts and the other a general fee increase with higher costs for both first and retake attempts, to explore how negative incentives affect students' academic persistence, course load, and performance. We find that penalizing module failures leads to changes in student behavior and performance, including reduced persistence and credit enrollment, with particularly large dropout effects among part-time students and those with lower academic performance, and a smaller additional effect among students from lower socio-economic backgrounds. Our results further show that financial pressure reduces the number of

attempts that students need to pass a module, which points to efficiency gains from retake penalties. In addition, cross-sectional analysis of graduation outcomes reveals that the raw gap in on-time graduation between self-funding and scholarship students is largely explained by pre-existing differences in academic preparation; conditional on prior performance, the fee policy has no significant additional effect on graduation among students who remain enrolled, suggesting that the negative impact on degree completion operates primarily through dropout.

Our evidence comes from the largest university in Catalonia, a public institution operating under the national scholarship rules that apply to all Spanish public universities. The findings are therefore informative for other publicly funded higher education systems with low baseline tuition, such as those in many European and Asian countries. A limitation of our study is that the empirical approach relies on a conditional parallel trends assumption, and the group-specific trends that we include may also absorb part of the post-treatment dynamics. We find similar results when we re-weight the comparison group by propensity-score matching instead of relying on these trends, and when we define treatment on pre-reform scholarship status only. Future work with data on household income and student employment could separate the roles of financial and time constraints, which our measures of socio-economic status and part-time study cannot distinguish.

Our results suggest that while financial penalties can be an effective tool for reducing inefficiencies in higher education, when implementing penalizing measures in higher education policy, broader goals of equitable access might be unintentionally undermined. To improve efficiency without exacerbating inequalities, policy makers should identify the students who are particularly vulnerable to such policies, such as part-time and academically marginal students, and include support mechanisms for them.

Tables and Graphs

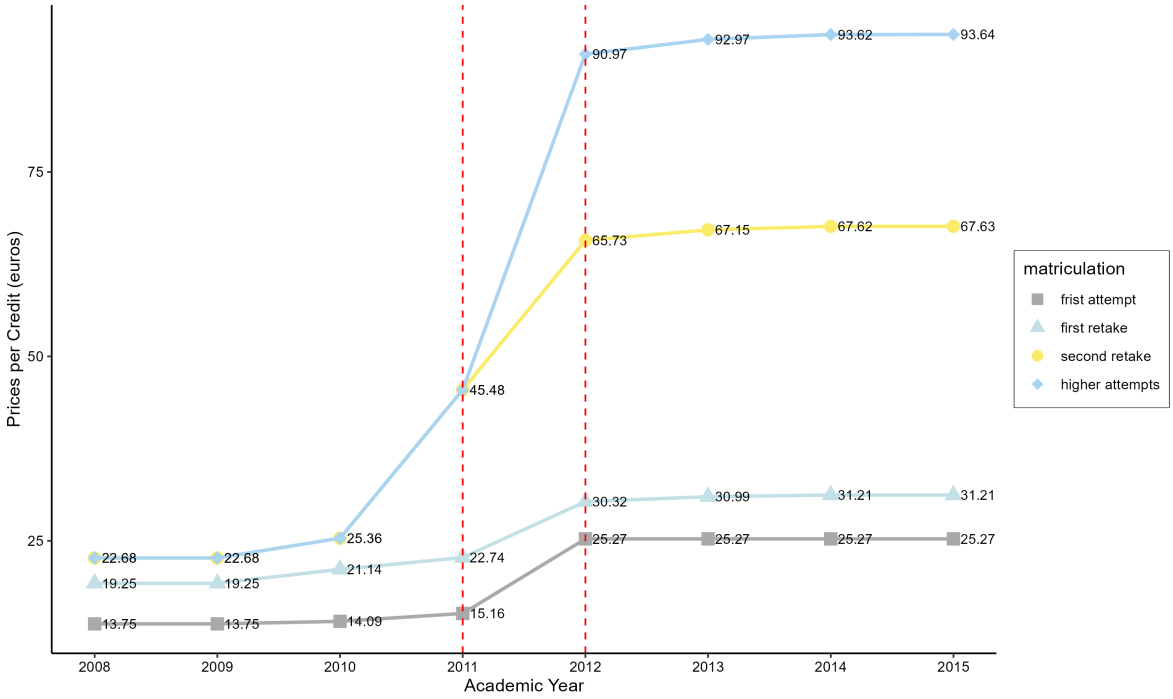


Figure 1: Evolution of Tuition Fees, Catalonia, 2008-2015

Notes: This figure presents the evolution of tuition fees in public universities in Catalonia during 2008 and 2015, with the fee levels of humanities and social sciences as an example. Fee levels for natural sciences, health sciences, and engineering are in general higher, but the evolution patterns are identical to the one presented in this graph.

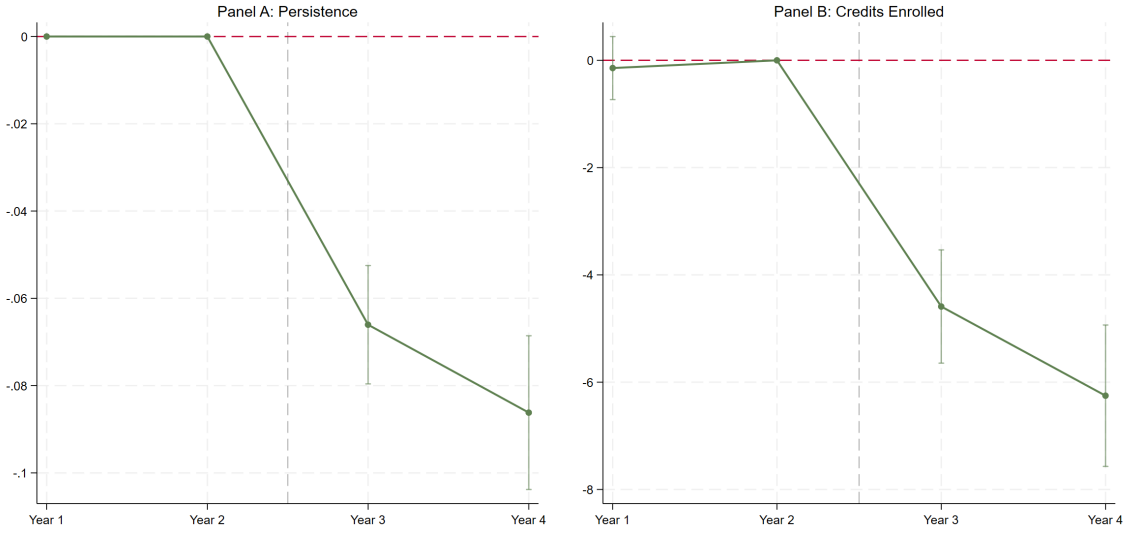


Figure 2: Student Level Impacts: Persistence and Credits Enrolled

Notes: Each panel plots the event-study coefficients from our preferred specification, which includes individual fixed effects, group-specific time trends on background characteristics, and group-specific trends on pre-treatment performance. The omitted base period is Year 2 (pre-treatment). The vertical dashed line marks the policy change point between Year 2 and Year 3. Error bars represent 95% confidence intervals. Standard errors are clustered at the individual student level. For persistence (Panel A), the Year 1 coefficient is mechanically zero with no CI as all students are active in Years 1 and 2 by sample construction.

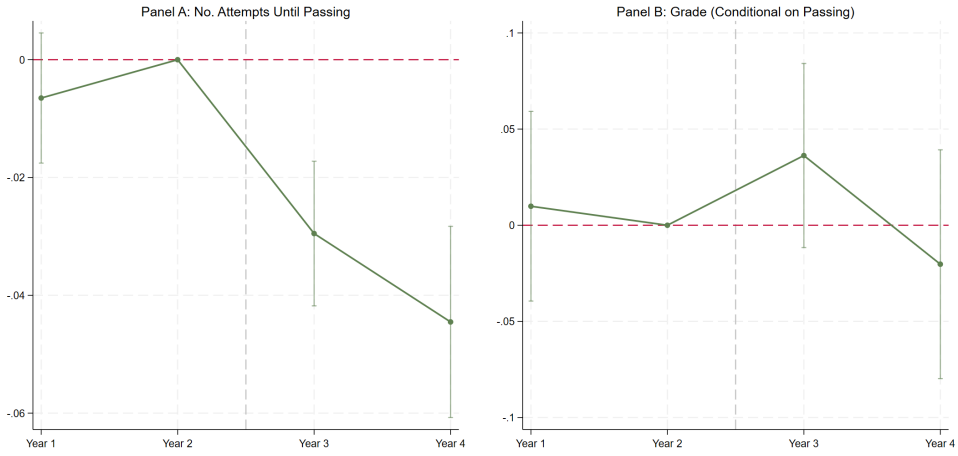


Figure 3: Module Level Impacts: Attempts and Grades

Notes: Panel A plots the event-study coefficients for the number of attempts until passing; Panel B for the numeric grade (0–10 scale) conditional on passing. Sample restricted to compulsory modules. Preferred specification includes individual and module fixed effects, group-specific trends on background characteristics, and a pre-treatment GPA trend. Base period: Year 2. Error bars represent 95% confidence intervals; standard errors clustered at the individual level.

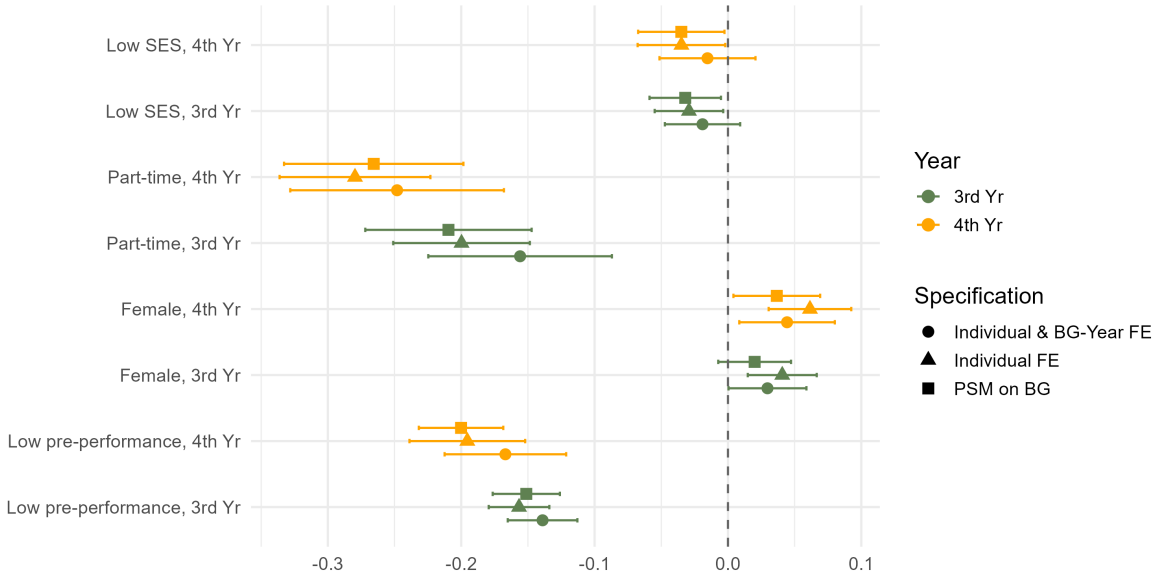


Figure 4: Heterogeneous Effects on Persistence

Notes: Each row plots the interaction coefficient of the heterogeneity specification, with results drawn from Table A13, indicating the additional effect of the fee increase for the indicated group relative to the non-group baseline. Green dots correspond to the third year of study (retake penalty wave); orange dots correspond to the fourth year (general fee increase wave). Three specifications are shown per group: Individual FE, Individual & BG-Year FE, and PSM on background characteristics; all include subgroup-specific year effects. Error bars represent 95% confidence intervals, with standard errors clustered at the student level.

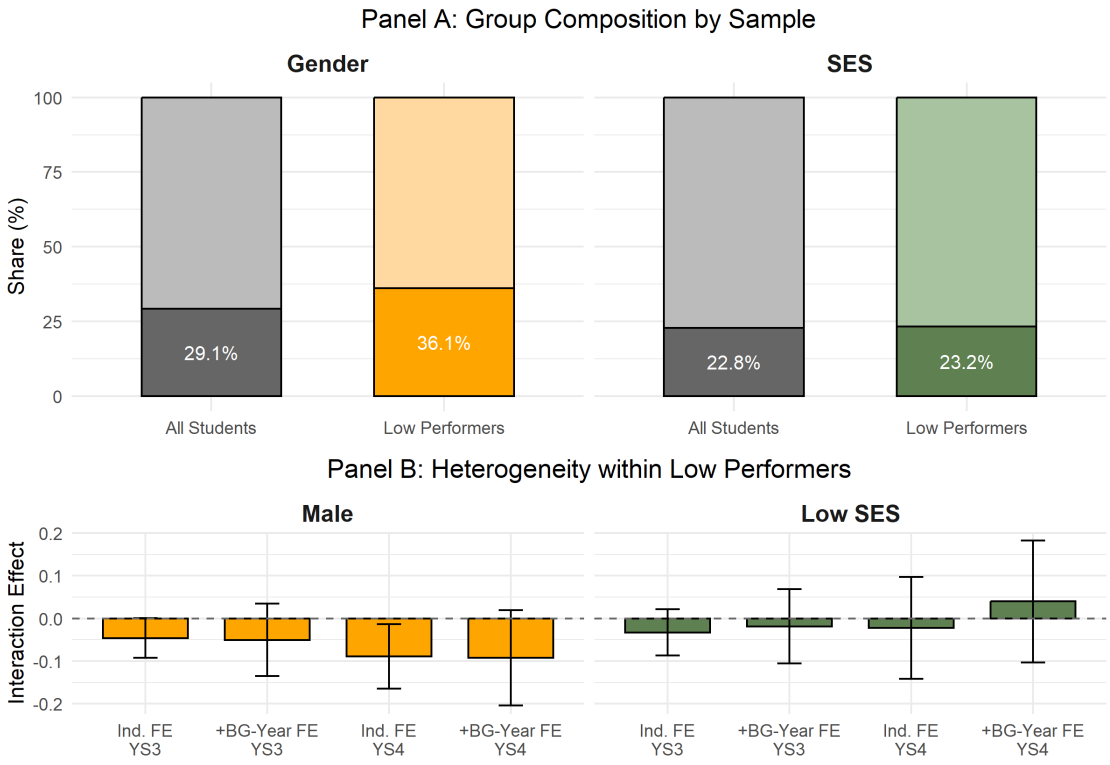


Figure 5: Mechanisms Driving Heterogeneity among Low Performers

Notes: Panel A shows the composition of the low-performer subgroup (students with pre-treatment GPA below 6) versus the full sample, broken down by SES and gender. Shares are descriptive statistics from the analytical sample. Panel B shows the additional dropout effect (interaction term) within the low-performer subsample for male students (left) and low-SES students (right), across two specifications (Individual FE and Individual & BG-Year FE) and two treatment periods (YS3 and YS4). Error bars represent 95% confidence intervals. Standard errors clustered at the student level.

Table 1: Scholarship Status Transition Matrix

		Year 1 → 2		Year 2 → 3		Year 3 → 4	
From ↓ / To →		Self-funded	Fee waiver	Self-funded	Fee waiver	Self-funded	Fee waiver
Self-funding	N	2,522	205	2,643	197	2,691	182
	%	92.5	7.5	93.1	6.9	93.7	6.3
Fee waiver	N	318	769	230	744	193	748
	%	29.3	70.7	23.6	76.4	20.5	79.5
Always same status		74.1% of sample (N = 2,825)					
Switchers		25.9% of sample (N = 989)					

Notes: This table reports year-to-year transitions in fee-waiver status for the analytical sample (N = 3,814 students). Each panel shows the number and row percentage of students moving between self-funding and fee-waiver status from one year of study to the next. The treatment definition (based on the first three years) is time-invariant and is not affected by Year 4 transitions.

Table 2: Summary Statistics

	With Fee Waiver		Without Fee Waiver		Norm.
	N	Mean/(SD)	N	Mean/(SD)	Diff.
Panel A: Background Characteristics					
Age at Enrollment	603	18.657	3,211	18.518	0.067
		(2.202)		(1.898)	
Female	603	0.796	3,211	0.692	0.239
		(0.403)		(0.462)	
Higher Education (Father)	603	0.055	3,210	0.257	-0.580
		(0.228)		(0.437)	
Higher Education (Mother)	603	0.063	3,210	0.244	-0.518
		(0.243)		(0.429)	
Unskilled Worker (Father)	603	0.063	3,210	0.031	0.151
		(0.243)		(0.174)	
Unskilled Worker (Mother)	603	0.133	3,210	0.073	0.198
		(0.339)		(0.260)	
Unemployed (Father)	603	0.005	3,210	0.001	0.074
		(0.070)		(0.031)	
Unemployed (Mother)	603	0.033	3,210	0.019	0.091
		(0.179)		(0.135)	
Foreigner	603	0.095	3,211	0.046	0.190
		(0.293)		(0.210)	
Admission Grades	603	6915.257	3,211	7005.948	-0.093
		(887.390)		(1053.409)	
Panel B: Student Outcomes					
Persistence	2,412	0.998	12,848	0.950	0.307
		(0.041)		(0.217)	
No. Credits Enrolled	2,412	60.190	12,848	56.876	0.278
		(5.829)		(15.803)	
Fees Paid (EUR)	2,409	124.987	12,192	1052.874	-1.786
		(296.497)		(672.081)	
No. Students		603		3,211	
No. Student-Year Records		2,412		12,848	

Notes: This table presents descriptive statistics for our analytic sample of the 2009 cohort, covering academic years 2009–2012. Panel A reports background characteristics at the student level and counts each student once; one student lacks information on parental background. Panel B reports outcomes at the student-year level for 3,815 student-program units, as one student is enrolled in two degree programs. Standard deviations in parentheses. The last column reports normalized differences, computed as $(\bar{X}_0 - \bar{X}_1)/\sqrt{(S_0^2 + S_1^2)/2}$, where group 0 refers to students with a fee waiver and group 1 to students without; values exceeding 0.25 in absolute value suggest meaningful imbalance.

Table 3: Effects on Persistence

	(1)	(2)	(3)	(4)
3rd Yr Study	-0.084***	-0.084***	-0.066***	-0.066***
	(0.005)	(0.006)	(0.007)	(0.007)
4th Yr Study	-0.108***	-0.108***	-0.086***	-0.086***
	(0.007)	(0.008)	(0.009)	(0.009)
N	15260	15260	15256	15256
Individual Fixed Effects		✓	✓	✓
Trends on Background			✓	✓
Trends on Pre-performance				✓
Unconditional Mean		0.958		

Notes: This table reports estimates of the effects on persistence for a variety of models. The sample includes students enrolled in the first two years of study, so persistence equals one in both years and no first-year estimate is reported; the omitted base period is the second year of study. Because the panel is balanced and treatment status does not vary over time, adding student fixed effects in Column (2) leaves the point estimates of Column (1) unchanged and only affects the standard errors. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student fixed effects. Column (3) incorporates group-specific time trends based on background characteristics including parental education and occupation, the student's admission grades to the program of study, student's English and French language abilities, major, gender, and whether the student is a foreigner. Column (4) further includes group-specific trends on students' performance before the increase, measured as the GPA the year before the shock. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects on Number of Credits Enrolled

	(1)	(2)	(3)	(4)
1st Yr Study	0.239	0.239	-0.079	-0.144
	(0.242)	(0.280)	(0.304)	(0.300)
3rd Yr Study	-4.778***	-4.778***	-4.545***	-4.590***
	(0.444)	(0.513)	(0.550)	(0.539)
4th Yr Study	-7.268***	-7.268***	-6.195***	-6.252***
	(0.525)	(0.606)	(0.694)	(0.672)
N	15260	15260	15256	15256
Individual Fixed Effects		✓	✓	✓
Trends on Background			✓	✓
Trends on Pre-performance				✓
Pre-treatment Mean (1st and 2nd Years)		60.416		

Notes: This table reports estimates of the effects on the number of credits enrolled for a variety of models. Students who are no longer enrolled are included with zero credits, and the omitted base period is the second year of study. Because the panel is balanced and treatment status does not vary over time, adding student fixed effects in Column (2) leaves the point estimates of Column (1) unchanged and only affects the standard errors. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student fixed effects. Column (3) incorporates group-specific time trends based on background characteristics including parental education and occupation, the student's admission grades to the program of study, student's English and French language abilities, major, gender, and whether the student is a foreigner. Column (4) further includes group-specific trends on students' performance before the increase, measured as the GPA the year before the shock. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effects on Passing

	(1)	(2)	(3)	(4)
Panel A: Number of attempts until passing				
1st Yr Study	-0.071***	-0.020***	-0.020***	-0.007
	(0.009)	(0.006)	(0.006)	(0.006)
3rd Yr Study	0.018*	-0.071***	-0.074***	-0.030***
	(0.011)	(0.006)	(0.006)	(0.006)
4th Yr Study	-0.047***	-0.104***	-0.103***	-0.045***
	(0.013)	(0.008)	(0.009)	(0.008)
Pre-treatment Mean (1st and 2nd Years)		1.199		
Panel B: Probability of passing within two attempts				
1st Yr Study	0.014***	0.003	0.004*	0.003
	(0.002)	(0.002)	(0.002)	(0.002)
3rd Yr Study	-0.014***	0.017***	0.018***	0.007***
	(0.003)	(0.002)	(0.002)	(0.002)
4th Yr Study	-0.003	0.015***	0.014***	0.007***
	(0.004)	(0.002)	(0.002)	(0.002)
Pre-treatment Mean (1st and 2nd Years)		0.965		
N	112556	112451	112451	112451
Individual and Module Fixed Effects		✓	✓	✓
Group Trends on Background			✓	✓
Group Trends on Pre-performance				✓

Notes: This table reports estimates of the effects on passing compulsory modules for a variety of models, where Panel A shows the effects on the number of attempts until passing, and Panel B shows the effects on the probability of passing within two attempts. Observations are at the student-module level for compulsory modules, by year of study, and the omitted base period is the second year of study. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student and module fixed effects. Column (3) incorporates group-specific time trends based on background characteristics including parental education and occupation, the student's admission grades to the program of study, student's English and French language abilities, major, gender, and whether the student is a foreigner. Column (4) further includes group-specific trends on students' performance before the increase, measured as the GPA the year before the shock. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Effects on Grades after Passing

	(1)	(2)	(3)	(4)
1st Yr Study	0.039	0.042*	0.036	0.010
	(0.043)	(0.024)	(0.025)	(0.025)
3rd Yr Study	0.005	0.076***	0.073***	0.036
	(0.042)	(0.024)	(0.025)	(0.024)
4th Yr Study	-0.177***	0.041	0.032	-0.020
	(0.050)	(0.030)	(0.031)	(0.030)
Pre-treatment Mean (1st and 2nd Years)		6.973		
N	100003	99874	99874	99874
Individual and Module Fixed Effects		✓	✓	✓
Group Trends on Background			✓	✓
Group Trends on Pre-performance				✓

Notes: This table presents the effects on grades for compulsory modules conditional on passing the module for a variety of models. Grades are on the 0–10 scale, with 5 as the passing grade. Observations are at the student-module level, by year of study, and the omitted base period is the second year of study. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student and module fixed effects. Column (3) incorporates group-specific time trends based on background characteristics including parental education and occupation, the student's admission grades to the program of study, student's English and French language abilities, major, gender, and whether the student is a foreigner. Column (4) further includes group-specific trends on students' performance before the increase, measured as the GPA the year before the shock. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Appendix A: Additional Results

Table A1: Effects on Persistence (PSM Sample)

	(1)	(2)	(3)	(4)	(5)
3rd Yr Study	-0.080***	-0.080***	-0.067***	-0.067***	-0.080***
	(0.005)	(0.006)	(0.007)	(0.007)	(0.006)
4th Yr Study	-0.104***	-0.104***	-0.087***	-0.087***	-0.107***
	(0.007)	(0.008)	(0.009)	(0.009)	(0.007)
N	14224	14224	14224	14224	14224
Individual Fixed Effects		✓	✓	✓	✓
Trends on Background			✓	✓	
Trends on Pre-performance				✓	
Matching					✓

Notes: This table reports estimates of the effects on persistence for a variety of models with the sample on the common support in the propensity score matching with the Epanechnikov kernel. Propensity scores are estimated with a probit on the background characteristics used in Column (3), with major entered as a set of indicators. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student fixed effects. Column (3) incorporates group-specific time trends based on background characteristics including parental education and occupation, the student's admission grades to the program of study, student's English and French language abilities, major, gender, and whether the student is a foreigner. Column (4) further includes group-specific trends on students' performance before the increase, measured as the GPA the year before the shock. Column (5) re-weights the sample using the PSM weights and controls for student fixed effects. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Effects on Persistence and Credits Enrolled (Matching Estimators)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Persistence								
3rd Yr Study	-0.080***	-0.080***	-0.080***	-0.080***	-0.080***	-0.080***	-0.080***	-0.080***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
4th Yr Study	-0.107***	-0.107***	-0.111***	-0.107***	-0.108***	-0.107***	-0.111***	-0.107***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
N	14224	14240	12303	14224	14184	14236	13253	13834
Panel B: Number of credits enrolled								
1st Yr Study	0.631	0.573	-1.260***	0.692	0.561	0.575*	0.280	0.329
	(0.443)	(0.359)	(0.186)	(0.460)	(0.455)	(0.349)	(0.478)	(0.487)
3rd Yr Study	-5.565***	-5.545***	-5.748***	-5.545***	-5.752***	-5.528***	-5.915***	-5.915***
	(0.623)	(0.569)	(0.387)	(0.620)	(0.641)	(0.578)	(0.632)	(0.630)
4th Yr Study	-6.786***	-7.147***	-9.809***	-6.845***	-6.339***	-7.283***	-8.269***	-6.709***
	(0.961)	(0.769)	(0.442)	(0.918)	(1.287)	(0.746)	(0.620)	(1.189)
N	14224	14240	12303	14224	14184	14236	13253	13834
Individual Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Matching Algorithm	Epanechnikov	Gaussian	Biweight	Radius (0.05)	Radius (0.01)	Radius (0.1)	3-Nearest	5-Nearest

Notes: This table presents robustness checks for the results in Tables 3 and 4, showing the effects on persistence (Panel A) and on the number of credits enrolled (Panel B) with student fixed effects, after employing various propensity score matching algorithms. Propensity scores are estimated with a probit on background characteristics, with major entered as a set of indicators. Columns (1)-(3) display results with kernel matching, using the Epanechnikov kernel, Gaussian kernel, and biweight kernel, respectively. Columns (4)-(6) display results with radius matching, using a radius of 0.05, 0.01, and 0.1, respectively. Columns (7)-(8) display results with 3- and 5-Nearest neighbors matching algorithms. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Effects on Number of Credits Enrolled (PSM Sample)

	(1)	(2)	(3)	(4)	(5)
1st Yr Study	0.152	0.152	-0.050	-0.146	0.631
	(0.246)	(0.284)	(0.304)	(0.301)	(0.443)
3rd Yr Study	-5.212***	-5.212***	-4.591***	-4.657***	-5.565***
	(0.421)	(0.486)	(0.547)	(0.535)	(0.623)
4th Yr Study	-7.213***	-7.213***	-6.212***	-6.299***	-6.786***
	(0.535)	(0.617)	(0.698)	(0.675)	(0.961)
N	14224	14224	14224	14224	14224
Individual Fixed Effects		✓	✓	✓	✓
Trends on Background			✓	✓	
Trends on Pre-performance				✓	
Matching					✓

Notes: This table reports estimates of the effects on numbers of credits enrolled for a variety of models with the sample on the common support in the propensity score matching with the Epanechnikov kernel. Propensity scores are estimated with a probit on the background characteristics used in Column (3), with major entered as a set of indicators. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student fixed effects. Column (3) incorporates group-specific time trends based on background characteristics including parental education and occupation, the student's admission grades to the program of study, student's English and French language abilities, major, gender, and whether the student is a foreigner. Column (4) further includes group-specific trends on students' performance before the increase, measured as the GPA the year before the shock. Column (5) applies propensity score matching with the Epanechnikov kernel before implementing the dynamic DID and includes individual fixed effects. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Effects on Passing (All Records)

	(1)	(2)	(3)	(4)
Panel A: Number of attempts until passing				
1st Yr Study	-0.072***	-0.021***	-0.022***	-0.007
	(0.009)	(0.006)	(0.006)	(0.006)
3rd Yr Study	0.014	-0.071***	-0.074***	-0.029***
	(0.011)	(0.006)	(0.006)	(0.006)
4th Yr Study	-0.048***	-0.105***	-0.104***	-0.045***
	(0.013)	(0.008)	(0.009)	(0.008)
Pre-treatment Mean (1st and 2nd Years)		1.200		
Panel B: Probability of passing within two attempts				
1st Yr Study	0.015***	0.003	0.004**	0.003
	(0.002)	(0.002)	(0.002)	(0.002)
3rd Yr Study	-0.013***	0.017***	0.018***	0.007***
	(0.003)	(0.002)	(0.002)	(0.002)
4th Yr Study	-0.003	0.015***	0.014***	0.007***
	(0.004)	(0.002)	(0.002)	(0.002)
Pre-treatment Mean (1st and 2nd Years)		0.965		
N	113205	113106	113106	113106
Individual and Module Fixed Effects		✓	✓	✓
Group Trends on Background			✓	✓
Group Trends on Pre-performance				✓

Notes: This table reports estimates of the effects on passing compulsory modules for a variety of models, including the records of cancelled enrollments and of credits obtained through recognition or validation, where Panel A shows the effects on the number of attempts until passing, and Panel B shows the effects on the probability of passing within two attempts. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student and module fixed effects. Column (3) incorporates group-specific time trends based on background characteristics. Column (4) includes both group-specific trends and pre-treatment outcomes. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Effects on Grades after Passing (All Records)

	(1)	(2)	(3)	(4)
1st Yr Study	0.039	0.042*	0.036	0.010
	(0.043)	(0.024)	(0.025)	(0.025)
3rd Yr Study	0.001	0.076***	0.073***	0.036
	(0.042)	(0.024)	(0.025)	(0.024)
4th Yr Study	-0.177***	0.041	0.032	-0.020
	(0.050)	(0.030)	(0.031)	(0.030)
Pre-treatment Mean (1st and 2nd Years)		6.973		
N	100376	100255	100255	100255
Individual and Module Fixed Effects		✓	✓	✓
Group Trends on Background			✓	✓
Group Trends on Pre-performance				✓

Notes: This table presents the effects on grades for compulsory modules conditional on passing the module for a variety of models, including the records of cancelled enrollments and of credits obtained through recognition or validation. Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student and module fixed effects. Column (3) incorporates group-specific time trends based on background characteristics. Column (4) includes both group-specific trends and pre-treatment outcomes. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Effects on Persistence (Pre-Reform Scholarship Status)

	(1)	(2)	(3)	(4)
3rd Yr Study	-0.054***	-0.054***	-0.045***	-0.045***
	(0.008)	(0.009)	(0.009)	(0.009)
4th Yr Study	-0.077***	-0.077***	-0.068***	-0.068***
	(0.009)	(0.010)	(0.011)	(0.011)
N	15260	15260	15256	15256
Individual Fixed Effects		✓	✓	✓
Trends on Background			✓	✓
Trends on Pre-performance				✓
Unconditional Mean		0.958		

Notes: This table reports robustness checks using an alternative treatment definition based only on scholarship status before the policy change: the control group consists of students who held a fee waiver in both the first and the second year of study, and the treated group of all other students. Unlike the main definition, it does not require control students to be enrolled in the third year. All other specifications are identical to Table 3. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Effects on Credits Enrolled (Pre-Reform Scholarship Status)

	(1)	(2)	(3)	(4)
1st Yr Study	0.610**	0.610**	0.416	-0.055
	(0.251)	(0.290)	(0.308)	(0.287)
3rd Yr Study	-3.960***	-3.960***	-2.684***	-3.013***
	(0.721)	(0.832)	(0.734)	(0.719)
4th Yr Study	-4.371***	-4.371***	-3.843***	-4.263***
	(0.644)	(0.743)	(0.797)	(0.771)
N	15260	15260	15256	15256
Individual Fixed Effects		✓	✓	✓
Trends on Background			✓	✓
Trends on Pre-performance				✓
Pre-treatment Mean (1st and 2nd Years)		60.416		

Notes: This table reports robustness checks using an alternative treatment definition based only on scholarship status before the policy change: the control group consists of students who held a fee waiver in both the first and the second year of study, and the treated group of all other students. Unlike the main definition, it does not require control students to be enrolled in the third year. All other specifications are identical to Table 4. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: First Stage: Effects on Total Fees Paid

	(1)	(2)	(3)	(4)
1st Yr Study	-239.767***	-239.767***	-175.158***	-118.816***
	(9.343)	(10.788)	(15.554)	(26.412)
3rd Yr Study	-10.917	-10.917	13.422	371.921***
	(10.723)	(12.382)	(16.820)	(32.894)
4th Yr Study	161.481***	161.481***	124.028***	362.539***
	(27.473)	(31.722)	(36.104)	(54.011)
N	15260	15260	15256	15256
Individual Fixed Effects		✓	✓	✓
Trends on Background			✓	✓
Trends on Pre-performance				✓
Pre-treatment Mean (1st and 2nd Years)		716.8		

Notes: This table reports estimates of the first-stage effect on total fees paid (IMTOTP, in EUR). The treated group consists of self-funding students (without fee waiver), who are fully exposed to the retake fee penalty; the control group consists of scholarship holders, whose fees are waived. The coefficients capture how the fee gap between the two groups changes relative to the base period (Year 2). In Year 4, when the general fee increase applied to all enrollments, the coefficient is positive and significant in all specifications. In Year 3, when only second and subsequent retakes became more expensive, the change in total fees paid is small and not statistically significant in Columns (1)–(3), in part because treated students also enrolled in fewer credits, and positive and significant in Column (4). Column (1) reports the results from a basic dynamic DID specification. Column (2) adds student fixed effects. Column (3) incorporates group-specific time trends based on background characteristics. Column (4) further includes group-specific trends on students' pre-treatment performance. Standard errors reported in parentheses are clustered at the individual student level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Graduation Outcomes: On-time Graduation (Full Sample)

	(1)	(2)	(3)	(4)
Self-funding (Treated)	-0.304***	-0.310***	-0.189***	-0.052***
	(0.015)	(0.016)	(0.015)	(0.013)
N	3814	3813	3813	3813
Background Controls		✓	✓	✓
Major Fixed Effects			✓	✓
Pre-treatment Performance				✓
Control Group Mean		0.891		

Notes: Cross-sectional regression of on-time graduation (graduated within 4 years) on treatment status (self-funding = 1). Students who never graduated are coded as 0. The treated group consists of self-funding students who are exposed to the retake fee penalty; the control group consists of scholarship holders who are shielded from retake fees. Specification (1) is unconditional; (2) adds background controls (age, gender, parental education, parental employment, nationality, university entrance exam score); (3) adds major fixed effects; (4) further adds pre-treatment academic performance (GPA, total credits, pass rate, and exam absence ratio). Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Graduation Outcomes: On-time Graduation (4-Year Stayers)

	(1)	(2)	(3)	(4)
Self-funding (Treated)	-0.233***	-0.233***	-0.135***	-0.018
	(0.015)	(0.016)	(0.015)	(0.013)
N	3429	3428	3428	3428
Background Controls		✓	✓	✓
Major Fixed Effects			✓	✓
Pre-treatment Performance				✓
Control Group Mean		0.896		

Notes: Cross-sectional regression restricted to students who remained actively enrolled in all 4 academic years ("stayers"). On-time graduation is coded as 1 if the student graduated within 4 years and 0 otherwise. The treated group consists of self-funding students; the control group consists of scholarship holders. Specifications follow the same structure as Table A9. The non-significant coefficient in Specification (4) indicates that, conditional on pre-treatment academic performance, stayers in the treated group graduate on time at rates similar to those of the control group. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Lee (2009) Bounds for Sample Selection

	Year of Study 3				Year of Study 4			
	Estimate	Lee Bounds		Trim.	Estimate	Lee Bounds		Trim.
		Lower	Upper	(%)		Lower	Upper	(%)
Panel A: student-years with enrolment								
Credits enrolled	−0.75	−1.74	0.10	8.4	−1.47	−3.22	−0.48	10.9
		[−2.35	0.77]			[−3.92	0.39]	
Panel B: compulsory modules (sample of Tables 5 and 6)								
Attempts until passing	−0.030	−0.033	−0.010	11.3	−0.045	−0.053	−0.017	11.5
		[−0.046	0.000]			[−0.069	−0.003]	
Passing within two attempts	0.007	0.002	0.009	11.3	0.007	0.000	0.009	11.5
		[−0.002	0.013]			[−0.003	0.013]	
Grade, conditional on passing	0.036	−0.064	0.147	11.8	−0.020	−0.154	0.116	12.5
		[−0.109	0.190]			[−0.209	0.171]	

Notes: This table reports Lee (2009) bounds for the effects on outcomes that are only observed for students who remain enrolled. For each outcome and year of study, the trimming share (Trim.) equals one minus the ratio of the share of treated students with an observed outcome to the corresponding share among control students. Control students with an observed outcome are ranked by the change in the outcome between the second year and the given year; the lower (upper) bound re-estimates the preferred specification (Column 4 of Tables 4–6) after dropping the bottom (top) trimming share of these students, and "Estimate" reports the untrimmed estimate. Panel A uses the student-years in which the student is enrolled. Panel B uses the sample of Tables 5 and 6; for the ranking, module-level outcomes are taken net of module means and averaged by student and year. The bounds rely on the monotonicity assumption of Lee (2009): the fee increases may lead some students to leave, but lead no student to stay who would otherwise have left. Brackets report a conservative 95% interval, from the lower bound minus 1.96 standard errors to the upper bound plus 1.96 standard errors, with standard errors clustered at the individual student level.

Table A12: Balance of Background Characteristics with Normalized Differences

Variable	(1) Total N	(1) Total Mean/(SD)	(2) 0 N	(2) 0 Mean/(SD)	(3) 1 N	(3) 1 Mean/(SD)	(2)-(3) Pairwise t-test N	(2)-(3) Pairwise t-test Normalized difference
Age at Enrollment	3815	18.541	603	18.657	3212	18.519	3815	0.067
		(1.949)		(2.202)		(1.898)		
Female	3815	0.709	603	0.796	3212	0.692	3815	0.239***
		(0.454)		(0.403)		(0.462)		
Higher Education (Father)	3814	0.225	603	0.055	3211	0.257	3814	-0.580***
		(0.417)		(0.228)		(0.437)		
Higher Education (Mother)	3814	0.215	603	0.063	3211	0.244	3814	-0.517***
		(0.411)		(0.243)		(0.429)		
Unskilled Worker (Father)	3814	0.036	603	0.063	3211	0.031	3814	0.151***
		(0.187)		(0.243)		(0.174)		
Unskilled Worker (Mother)	3814	0.082	603	0.133	3211	0.073	3814	0.198***
		(0.275)		(0.339)		(0.260)		
Unemployed (Father)	3814	0.002	603	0.005	3211	0.001	3814	0.074**
		(0.040)		(0.070)		(0.031)		
Unemployed (Mother)	3814	0.021	603	0.033	3211	0.019	3814	0.091**
		(0.143)		(0.179)		(0.135)		
Foreigner	3815	0.054	603	0.095	3212	0.046	3815	0.190***
		(0.226)		(0.293)		(0.210)		
Admission Grades	3815	6991.240	603	6915.257	3212	7005.504	3815	-0.093**
		(1029.490)		(887.390)		(1053.546)		

Notes: This table reports means, standard deviations, and normalized differences (Imbens & Wooldridge, 2009) between the treated group (self-funding students exposed to retake fees) and the control group (scholarship holders shielded from retake fees) for pre-determined background characteristics. Normalized differences are computed as $(\bar{X}_0 - \bar{X}_1)/\sqrt{(S_0^2 + S_1^2)/2}$, where $\bar{X}_g$ and S_g^2 denote the sample mean and variance for group g, with group 0 the control group and group 1 the treated group. Values exceeding 0.25 in absolute value suggest meaningful imbalance (Imbens & Wooldridge, 2009). Counts refer to the 3,815 student-program units of the estimation sample; the one student enrolled in two degree programs is counted once in Table 2. HE = higher education.

Table A13: Heterogeneous Effects on Persistence

	Low SES			Part-time			Female			Low pre-performance		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
3rd Yr (0)	-0.075***	-0.059***	-0.068***	-0.060***	-0.054***	-0.057***	-0.112***	-0.089***	-0.093***	-0.012***	-0.017***	-0.011***
	(0.006)	(0.009)	(0.007)	(0.005)	(0.007)	(0.005)	(0.012)	(0.014)	(0.012)	(0.003)	(0.006)	(0.003)
Diff 3rd Yr	-0.029**	-0.019	-0.032**	-0.200***	-0.156***	-0.210***	0.041***	0.030**	0.020	-0.157***	-0.139***	-0.151***
	(0.013)	(0.014)	(0.014)	(0.026)	(0.035)	(0.032)	(0.013)	(0.015)	(0.014)	(0.012)	(0.013)	(0.013)
4th Yr (0)	-0.098***	-0.080***	-0.087***	-0.075***	-0.070***	-0.073***	-0.152***	-0.121***	-0.125***	-0.012***	-0.025***	-0.010*
	(0.009)	(0.012)	(0.010)	(0.007)	(0.009)	(0.007)	(0.013)	(0.016)	(0.014)	(0.005)	(0.007)	(0.006)
Diff 4th Yr	-0.035**	-0.015	-0.035**	-0.280***	-0.248***	-0.266***	0.061***	0.044**	0.037**	-0.195***	-0.167***	-0.200***
	(0.017)	(0.018)	(0.017)	(0.029)	(0.041)	(0.034)	(0.016)	(0.018)	(0.017)	(0.022)	(0.023)	(0.016)
Constant	1.000***	1.322***	1.000***	1.000***	1.242***	1.000***	1.000***	1.322***	1.000***	1.000***	1.118***	1.000***
	(0.002)	(0.069)	(0.002)	(0.002)	(0.071)	(0.002)	(0.002)	(0.068)	(0.002)	(0.002)	(0.066)	(0.002)
N	15260	15256	12852	15260	15256	12852	15260	15256	12852	15260	15256	12852
Individual Fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Group Trends on Background		✓			✓			✓			✓	

Notes: This table presents the heterogeneous effects on persistence. In each group, the rows "3rd Yr (0)" and "4th Yr (0)" report the effect for students outside the subgroup (β_s in Equation 3), and the rows "Diff 3rd Yr" and "Diff 4th Yr" report the additional effect for students in the subgroup (b_s). The columns are grouped based on various demographic characteristics. Columns (1)–(3) display results for the interaction with low socioeconomic status (Low SES=1), Columns (4)–(6) display results for the interaction with part-time study mode (Part-time=1), Columns (7)–(9) display results for the interaction with gender (Female=1), and Columns (10)–(12) display results for the interaction with low pre-treatment GPA in the second year of study (Low pre-performance=1). Each group includes three columns representing different models or specifications used in the analysis. The first column of every group presents the event study specification with student fixed effects; the second incorporates group-specific time trends based on background characteristics (excluding those used as group variables); the third applies propensity score matching on background variables with the Epanechnikov kernel before implementing dynamic DID analysis. Standard errors reported in parentheses are clustered at the individual student level. All specifications include subgroup-specific year-of-study effects. Full numerical results are visualized in Figure 4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B: Additional Information on Curriculum Change

In academic years 2009 and 2010, as part of the Bologna process, there were universal curriculum reforms happening in Catalonia. At the University of Barcelona, the introduction of the new curriculum started in 2009 and 2010, depending on the degrees. We refer to majors that started the new plan from 2009 as "early-changing" and those from 2010 as "late-changing". Students from cohorts before the change happened were allowed to change to the new plan without institutional costs borne from their side. This means that students in the 2009 cohort in these late-changing majors started with the old plan, but could change to the new one in 2010, their second year of study. As this involves a layer of student self-selection and introduces effects that cannot be separated from the effects of interest, we exclude these late-changing majors from the analysis. Below, we provide some details about this curriculum change, regarding teaching activities, exams, and enrollment.

Table A14 shows how modules designed for different years of study (1st, 2nd, 3rd, 4th) in both the new and old curricula were offered in the process of the curriculum change. The general idea is that the new curriculum (columns 1 and 2) should fade in with the old one (columns 3 and 4). The change of teaching activities was demonstrated in the first and third columns. In academic year 2009, the university provided first-year modules in the new curriculum for early-changing majors (late-changing majors had first- and second-year modules introduced at the same time in academic year 2010), as well as first- to fourth-year modules in the old curriculum. In the academic year of 2011, all modules in the new curriculum were already introduced. Meanwhile, the teaching of old modules faded out year by year, with academic year 2012 being the last year that a module in the old curriculum was taught. Exams for (and enrollment in) the modules were organized in a different way so that students on the old curriculum could still finish their studies if they did not choose to change curriculum. This means that the old ones faded out more slowly. Every module in the old curriculum was kept in the system for another two years after the last year of teaching activities. For example, in academic year 2011, students could still formally enroll in and take exams of their first-year modules if they failed previously, even though teaching was last offered two years ago.

The old and new curricula differ in many respects, even if the name of the major stays unchanged. First, a series of new modules was introduced, replacing old ones. Second, the new curriculum includes only compulsory and elective modules, all related to the major's field of study, whereas the old curriculum includes free-choice modules that cover a wider range of topics. Third, teaching methods also changed when shifting to the new plan, featuring lectures with smaller class sizes and tutorials. There are other aspects, such as the difference in the credit system and the introduction of continu-

ous assessments. Given the significant changes, ideally, it is better to examine the effects of tuition fee changes without being complicated by the change of curricula. For the 2009 cohort, students in early-changing majors participated in the learning activities under the new curriculum and did not experience the change, while students in the late-changing majors started with the old curriculum and many changed to the new one from their second year. Therefore, we restrict our sample by including only students studying the early-changing majors, as explained in Section 3.

Table A14: Process of Curriculum Change

Academic Year	New Plan (Teaching)	New Plan (Enrollment/Exams)	Old Plan (Teaching)	Old Plan (Enrollment/Exams)
2009/10	1st (early)	1st (early)	1st, 2nd, 3rd, 4th	1st, 2nd, 3rd, 4th
2010/11	1st (late), 2nd	1st (late), 2nd	2nd, 3rd, 4th	1st, 2nd, 3rd, 4th
2011/12	1st, 2nd, 3rd	1st, 2nd, 3rd	3rd, 4th	1st, 2nd, 3rd, 4th
2012/13	1st, 2nd, 3rd, 4th	1st, 2nd, 3rd, 4th	4th	2nd, 3rd, 4th
2013/14	1st, 2nd, 3rd, 4th	1st, 2nd, 3rd, 4th		3rd, 4th
2014/15		1st, 2nd, 3rd, 4th		4th
2015/16		1st, 2nd, 3rd, 4th		

Notes: This table presents the details of how modules designed for different years of study (1st, 2nd, 3rd, 4th) in both the new and old curricula were offered in the process of the curriculum change. In academic years 2009 and 2010, the timing of introducing new first-year modules differs for early-changing and late-changing majors, as shown in parentheses.